

# **Impact of Red Needle Cast defoliation of radiata pine stand on growth at Wharerata Forest - updated**

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**REPORT INFORMATION SHEET**

**REPORT TITLE** IMPACT OF RED NEEDLE CAST DEFOLIATION OF RADIATA PINE ON  
STAND GROWTH AT WHARERATA FOREST-UPDATED

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## EXECUTIVE SUMMARY

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The objective of the research described in this report was to determine the effect of defoliation owing to red needle cast (RNC) on growth rate of *Pinus radiata* stands at Wharerata Forest. Defoliation events due to RNC have been reported to occur at Wharerata in 2008 and 2011.

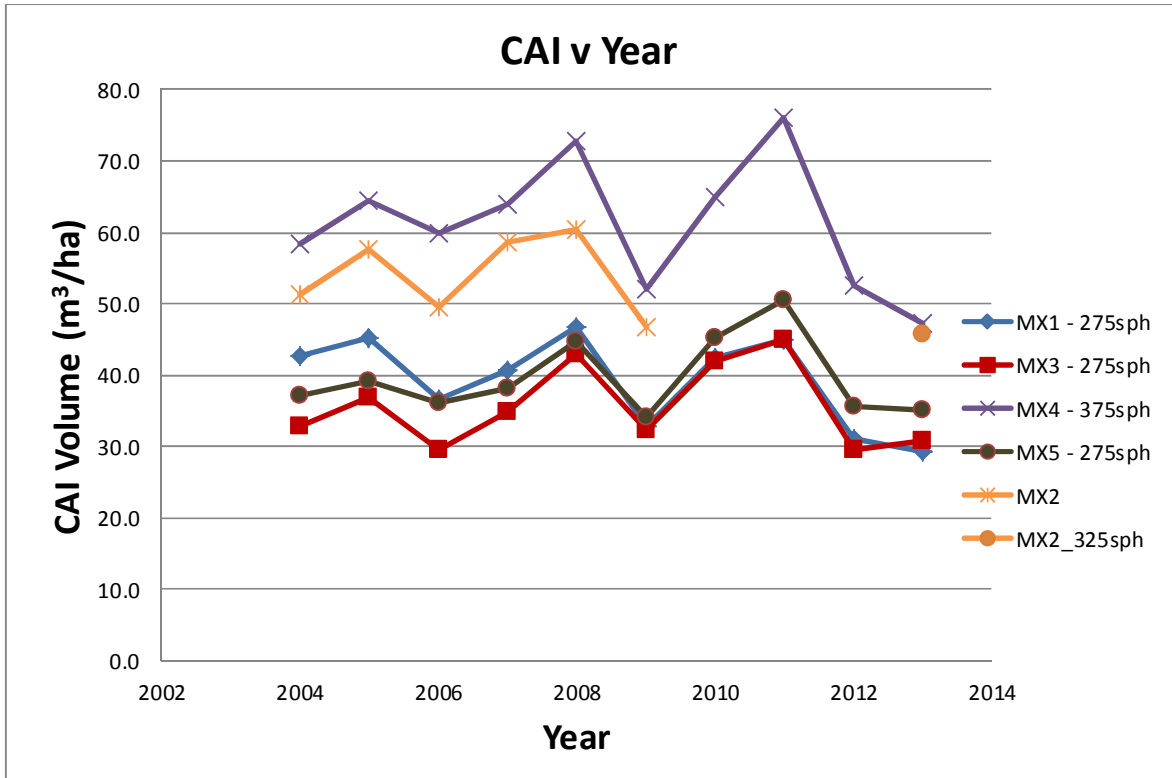
Five plots were installed at Maxwell Road in 22 year old stands known to be severely defoliated by red needle cast in the past, and five plots were installed at Akarana Road in 29 year old stands where moderate or no defoliation was evident in 2012. Tree total height, breast height diameter, and needle retention were measured in August 2012. Breast height cores were taken from all live plot trees. The width and density of annual growth rings were measured using a direct scanning wood densitometer. The cores showed that ring width initially increased with increasing tree age and then decreased, as part of the normal growth dynamics. However, there was an obvious reduction in stem cross-sectional area increment evident in 2009 and 2012 at Maxwell. Cross-sectional area increments decreased significantly at Maxwell Road in 2009 and 2012 relative to increments at Akarana Road plots at the corresponding stand ages.

This study provided statistically robust evidence that severe defoliation by RNC substantially reduced ring width and hence cross-sectional area increment. Defoliation events evidently occurred in 2008 and 2011, followed by a recovery in growth rate, presumably as the canopy leaf area rebuilt. The reduction in increment attained 38% in the growth year immediately following RNC infection, 10% in the subsequent year, with no loss thereafter until the next RNC event. In addition to influencing growth, RNC had a small effect on late wood density, which was reduced by RNC defoliation, and hence the range in wood density within rings decreased in rings affected by defoliation.

The effect of RNC on volume increment was estimated using the 300 Index growth model. The 300 index was calculated annually using the cross-sectional area data derived from the breast height cores, the predicted mean top height based on height measurements in 2012, and the silvicultural regime. An assessment of the trend in 300 Index over time suggested that there has been a small reduction in the 300 Index during the period when RNC became evident.

Plots at Maxwell Road were re-measured (dbh, needle retention) in 19<sup>th</sup> June 2013 to assess the current health of tree crowns and to determine current annual volume increment (CAI) from 2004-2013. The CAI in stem volume was derived each year using the 300 Index growth model. Model inputs required to calculate annual CAI's included the annual basal area and stocking data from cores to 2012, as described above, together with the basal area and stocking data acquired in 2013. The estimates of stem total volume (under bark) output annually from 2004-2013 were used to calculate the CAI in volume for each year from 2004 to 2013. The figure shows that peak defoliation events due to RNC in 2005, 2008, and 2011 were followed by low stem volume CAI's one year later in 2006, 2009, and 2012. Further, volume increment partially recovered two years later in 2007 (1142 mm rainfall) and 2010 (1632 mm rainfall), but not in 2013 (560 mm rainfall), even though average needle retention increased from 0.3 in 2012 to 1.2 years in 2013. We think that the expected improvement in volume increment did not occur in 2013 because of severe drought.

A re-measurement in 2014 should see a marked improvement in growth at Maxwell Road provided that defoliation due to RNC does not re-occur in Spring 2013 and there is adequate rainfall over the coming year.



The following sequence of events explains the patterns in ring width: 1) normal rainfall years lead to average growth rates, with narrow ring widths evident in low rainfall years; 2) RNC infection results in defoliation that year; 3) Defoliation markedly reduces growth the following year; 4) The crown and growth rate gradually recover until the sequence is repeated. Severe RNC events have occurred at Wharerata on a 3-year cycle. Each severe RNC event caused a 16% reduction in stem periodic mean annual cross sectional area increment. Overall growth loss will depend on the number and severity of defoliation events over the life of a stand.

# IMPACT OF RED NEEDLE CAST DEFOLIATION OF RADIATA PINE ON STAND GROWTH AT WHARERATA FOREST -UPDATED

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# 1. Introduction

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In the winter of 2008 and 2011 severe defoliation due to *Phytophthora taxon pluvialis*, or Red Needle Cast (RNC), occurred in *Pinus radiata* stands on the East Coast of New Zealand, where it has been noted since about 2005. RNC has also occurred intermittently in other regions, for example Northland, Coromandel, and central North Island. The proportion of the live crown defoliated varies from site to site, and from year to year. With RNC, all needle age classes die within the affected part of the crown. In severe cases needles of all age classes over almost the entire crown are affected, as occurred in 2011 at Maxwell Road, Wharerata where only a small proportion of trees appeared to be tolerant to RNC, with the majority of trees in the stand being almost completely defoliated.

Buds, and hence the spring/early autumn needle growth flush, appear to be largely unaffected by RNC, with more or less completely defoliated trees having an apparently full complement of current (0 – 1 year old) needles in the following summer. Tree crowns can be expected to recover over the next two years, due to needle retention, provided that RNC does not re-appear. Stem annual growth loss will therefore largely reflect the severity of defoliation during the growth year immediately following the defoliation event, followed by a recovery in stem growth over the next two years as the tree crowns recover.

No statistically robust estimates of the impact of RNC defoliation on radiata pine growth have been made. A study was therefore initiated in 2012 to quantify growth loss owing to RNC, as part of a larger research programme on RNC. The aim of this study was to quantify the effect of severe defoliation by RNC on stem cross-sectional area increment at Maxwell Road, where defoliation is known to have occurred in both 2008 and 2011. This study is intended to provide an indication of the likely maximum growth loss owing to each defoliation event.

The effect of RNC on annual cross-sectional area increment was determined from breast height cores. The effect on mean annual increment (MAI) of stem total volume under bark was estimated using the 300 Index growth model.

# 2. Materials and Methods

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## Field procedures

The approach involved installing and measuring plots (circular 0.04 ha horizontal area) in mid-rotation aged stands aged 22 years at Maxwell Road that had been intermittently affected by RNC, and uninfected control stands nearby. Five plots (MX1-5) were installed at Maxwell Road in stands known to be severely defoliated by red needle cast in 2008 and 2011, and five plots (AK6-10) were installed in older stands (aged 29 years) at Akarana Road. The plots at Maxwell and Akarana were installed and trees measured on 6<sup>th</sup> August 2012, when tree total height (Ht) and breast height diameter (dbh) were measured.

Plots at Maxwell Road were re-measured (dbh, needle retention) on the 19<sup>th</sup> June 2013, to assess the current health of tree crowns and to determine annual volume increment for the period from 2004-2013.

Needle retention was assessed per tree at Maxwell, and estimated for the stand at Akarana, where the trees were too tall to score individually.



Scores assigned to trees in August 2012 and 2013 were based on a count of the number of needle age classes present in the lower 1/3<sup>rd</sup> of the sunlit part of the crown:

- 0 = little or no live foliage present
- 1 = 1 needle age class
- 2 = 2 needle age classes
- 3 = 3 or more needle ages classes present

Mean needle retention scores (NR) in August 2012, one year following the severe RNC event in 2011, are given per plot in Table 1. A score of zero in August 2012 indicates that the buds in the lower 1/3<sup>rd</sup> of the crown had died and no new needles were therefore formed, while an average score of 1 in 2012 indicates that the lower 1/3<sup>rd</sup> of the crown of most trees had 1 year old needles (current foliage) present. The plots at Maxwell Road had average NR scores less than 1, which indicates that the lower 1/3<sup>rd</sup> of the crown of most trees failed to produce new needles. About half the plot trees had new foliage in the lower 1/3<sup>rd</sup> of the crown, except MX1 and MX5 where few trees had 1 year old needles in the lower 1/3<sup>rd</sup> of the crown. It is possible that some buds in the lower crown had died prior to the RNC event in 2011, although this seems unlikely because the canopies were rather open (See cover photograph for MX1 in 2012). Needle retention improved in 2013 by almost 1 year.



**Photo 1. Edge tree at Maxwell Road photographed in August 2012 and in June 2013, showing an improvement in needle retention score by approximately 1 year.**

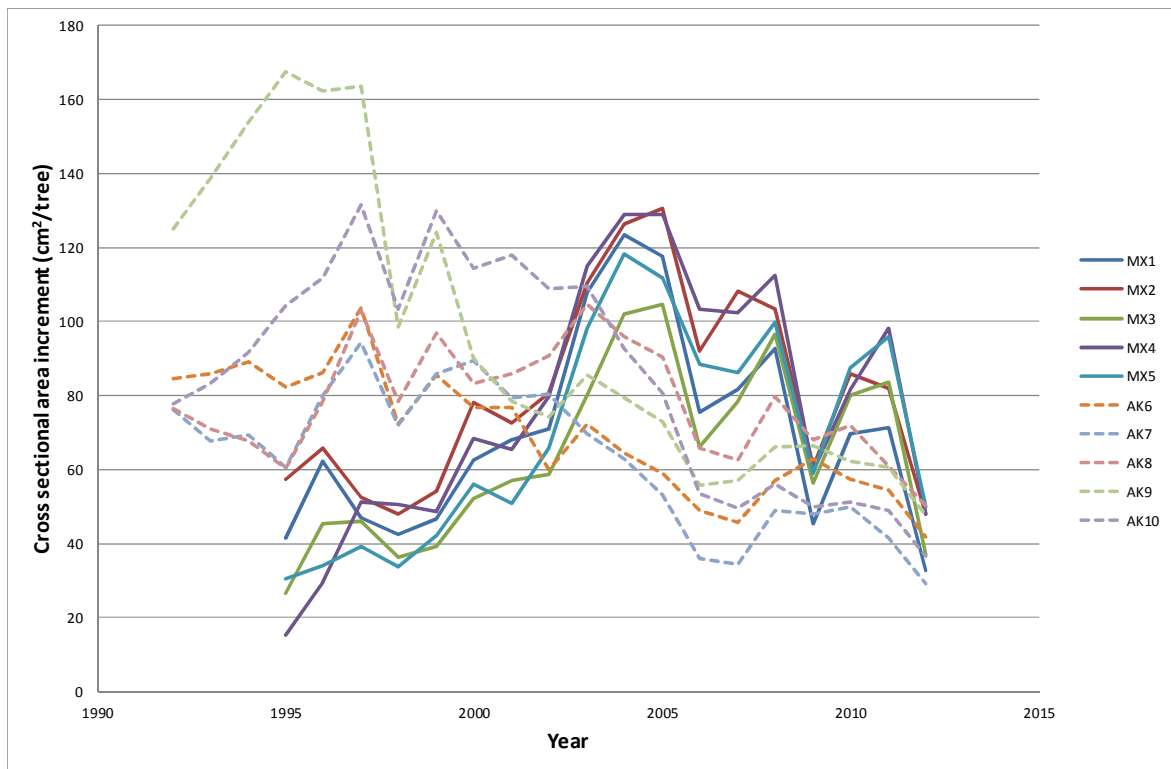
**TABLE 1. Plots installed and measured at Wharerata in August 2012 and mean needle retention (NR). The Akarana (control) plots were thought to be unaffected by RNC prior to 2006**

| <b>Plot</b> | <b>Easting</b> | <b>Northing</b> | <b>Elevation<br/>(m)</b> | <b>NR 2012 (yrs)</b> | <b>NR 2013 (yrs)</b> |
|-------------|----------------|-----------------|--------------------------|----------------------|----------------------|
| MX1         | E2929400       | N6243861        | 510                      | 0.1                  | 1.0                  |
| MX2         | E2929389       | N6243910        | 514                      | 0.6                  | 1.5                  |
| MX3         | E2929443       | N6244004        | 530                      | 0.4                  | 1.1                  |
| MX4         | E2929272       | N6243832        | 525                      | 0.4                  | 1.2                  |
| MX5         | E2929245       | N6243820        | 509                      | 0.2                  | 1.1                  |
| AK6         | E2930441       | N6241647        | 462                      | 1                    | Na                   |
| AK7         | E2930843       | N6241089        | 419                      | 1                    | Na                   |
| AK8         | E2930777       | N6241150        | 422                      | 1                    | Na                   |
| AK9         | E2931403       | N6241247        | 392                      | 1                    | Na                   |
| AK10        | E2931334       | N62410941       | 394                      | 2                    | Na                   |

The Akarana control plots were installed in apparently healthy stands and were of similar aspect and physiography (face, terrace) as the plots at Maxwell. However, only moderate needle retention was actually evident at Akarana (where the healthiest stands were found) in 2012, and some RNC symptoms were clearly evident. Therefore the Akarana control plots were deliberately installed in stands that were old enough to allow valid estimates of growth loss when the stands were of a comparable age as those at Maxwell Road.

A breast height 5mm pith-to-bark core sample was taken from all live plot trees. The width and density of annual growth rings from cores were measured using a direct scanning wood densitometer. Depth of core sampling allowed stem cross sectional area increment per plot to be calculated from either age 9 or age 5 years onwards at Akarana and Maxwell, respectively (Figure 1). The cores showed that average cross sectional area increment per tree for each plot initially increased with increasing tree age and then decreased, as part of the normal growth dynamics (Figure 1).





**Figure 1. Annual average cross sectional area increments of cores for all live trees per plot at Maxwell (Solid lines MX) and Akarana (dashed lines AK). For example, the annual increment from 1<sup>st</sup> July 1994 – 30<sup>th</sup> June 1995 is graphed as 1995.**

## Data analysis

To avoid the effects of silvicultural activities on annual cross sectional area increments, the impact of RNC on growth was analysed from stand age 15 years and older. The stands at Maxwell were 15 years old in 2005 and 22 years in 2012, whereas the Akarana stands were 15 years old in 1998 and 22 years in 2005 (Table 2). The analysis of growth loss by age as opposed to by year has the advantage that the Akarana controls were probably not exhibiting growth loss owing to RNC until 2006 at the earliest. However, there were differences in annual rainfall between years that need to be considered when analysing growth in relation to stand age. NIWA rainfall data from a site at Waingake (Table 2) were compiled for analysis of rainfall effects on growth. Based on these rainfall totals, there were probably only two years with insufficient rainfall (<1200 mm) to influence growth appreciably – in 1998 and 2007.

**Table 2. Annual rainfall totals at Waingake by growth year (1st July – 30th June) for years when stands at Maxwell and Akarana were the same age (Year and stand age at end of annual increment period)**

| Age | Maxwell (Planted 1990) |               | Akarana (Planted 1983) |               |
|-----|------------------------|---------------|------------------------|---------------|
|     | Year                   | Rainfall (mm) | Year                   | Rainfall (mm) |
| 15  | 2005                   | 1408.3        | 1998                   | 1117.6        |
| 16  | 2006                   | 2214.7        | 1999                   | 1396.1        |
| 17  | 2007                   | 1142.4        | 2000                   | 1307.9        |
| 18  | 2008                   | 1521.6        | 2001                   | 1289.5        |
| 19  | 2009                   | 1358.5        | 2002                   | 1271.6        |
| 20  | 2010                   | 1632          | 2003                   | 1551.7        |
| 21  | 2011                   | 1846.9        | 2004                   | 1913.1        |
| 22  | 2012                   | 1778.3        | 2005                   | 1408.3        |

RNC-related needle retention assessments were used to establish relationships between growth loss and RNC. To facilitate the analysis of growth loss it was decided to assign needle retention classes to years as follows. In the 2008-2009 and 2011-2012 growth year needle retention at Maxwell was classified as “Low”. In 2009-2010 growth year needle retention was classified as “Moderate”. It was assumed that other years at Maxwell and all years analysed at Akarana had “High” retention.

### **300 Index approach**

The annual cross sectional area measurements from cores allowed the 300 Index to be estimated annually from 2003 – 2012 per plot for all plots. Plots with a constant number of cored live trees were used, and the resulting trends in the 300 Index examined in relation to RNC infection. One plot at Maxwell and 2 plots at Akarana were not used because tree top outs (breakage) reduced stand growth in later years.

### **Statistical analysis**

The relationship between annual cross sectional area increments at Maxwell to that at Akarana was tested when annual growth rings were the same age. The GLM procedure in SAS was used to calculate the least squares mean cross sectional area increment by needle retention class. This modelling approach was repeated in an analysis involving only the plots from Maxwell, using annual growth rings formed from 1 July 2004 to 30 June 2005 and later.

## **3. Results and Discussion**

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### **General trends evident in the growth measurements**

Average cross-sectional area increment per tree is shown for each plot from age 9 years onwards in Figure 1, and from 2005 (for the 2004-2005 increment period) onwards in Figure 2, during which time no silvicultural events occurred. Increments in Figure 2 were generally larger in the younger plots at Maxwell (22 years in 2012) compared to Akarana (29 years in 2012), because of difference in stand age, site, and climate factors.

The strong year effects evident across plots were not entirely explained by annual rainfall totals, although low rainfall in 1998 and 2007 apparently reduced growth to some extent. For example, low increment was evident in all plots at both Maxwell and Akarana in 1998, which was a low rainfall year and before any RNC was known to be present. A marked decrease in cross-sectional area increment occurred at Maxwell Road in 2009 and again in 2012 which we ascribe to RNC. A recovery in increment occurred at Maxwell in the following year (2007, when rainfall was low but it would have been more effective because interception losses by the tree canopy would have been low). Cross-sectional area increment was at a minimum at Akarana in 2007, as expected owing to the low rainfall. It is noteworthy that increment was low at both sites in 2006, even though rainfall was high that year, which suggests that RNC reported to occur on the East Coast in 2005 was present at Wharerata Forest.

### **RNC effect on annual cross-sectional area increment, based on Maxwell and Akarana plots**

The analysis of data illustrated in Figure 1 showed that needle retention (prob. > F for NR = 0.0037) strongly influenced annual increment, after allowing for stand age effects (prob. >F for Age < 0.0001), with age and NR both fitted as class variables. It was necessary to fit age as a class variable because growth depended on various factors, including RNC. The least squares means for the three needle retention classes are given below:

| NR class | LS mean increment | 95% CI's    |
|----------|-------------------|-------------|
| High     | 72.2              | 68.9 – 75.5 |
| Medium   | 64.8              | 37.2 – 92.5 |
| Low      | 44.8              | 25.6 – 64.1 |

Growth loss is therefore 38%  $\{100 \times (72.2 - 44.8)/72.2\}$  one year after the infection and 10% two years after the infection. No growth loss (actually would be small but insignificant) was assumed to occur three years after the infection.

### **RNC effect on annual cross sectional increment based solely on Maxwell plots**

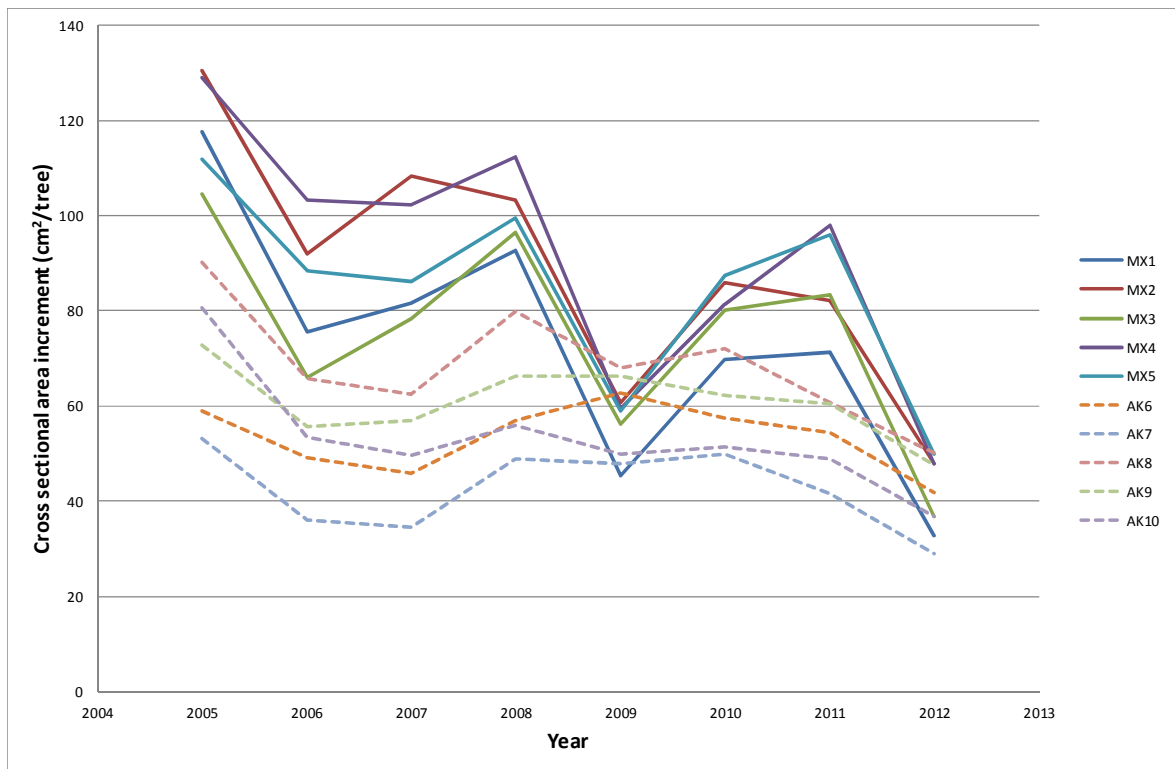
Restricting the statistical analysis of RNC effects to Maxwell plot data shown in Figure 2, the growth loss estimates in relation to NR, after allowing for age effects, were virtually the same as above, although the importance of NR increased. Age (prob.  $>F = 0.0044$ ) and NR (prob.  $> F > 0.0001$ ) both significantly influenced annual increment. Age was significant because cross sectional area increment at breast height tended to decrease with increasing age (as continuous variable) over this age range. The least squares means for the three needle retention classes are given below, with 95% confidence intervals CI indicating that the High/Medium class means are significantly distinct from the Low class mean:

| NR class | LS mean increment | 95% CI's     |
|----------|-------------------|--------------|
| High     | 92.6              | 85.4 – 99.9  |
| Medium   | 86.2              | 71.0 – 101.4 |
| Low      | 56.6              | 44.8 – 68.4  |

Growth loss is therefore 39% one year after the infection and 8% two years after the infection, with no growth loss assumed three years after the infection.

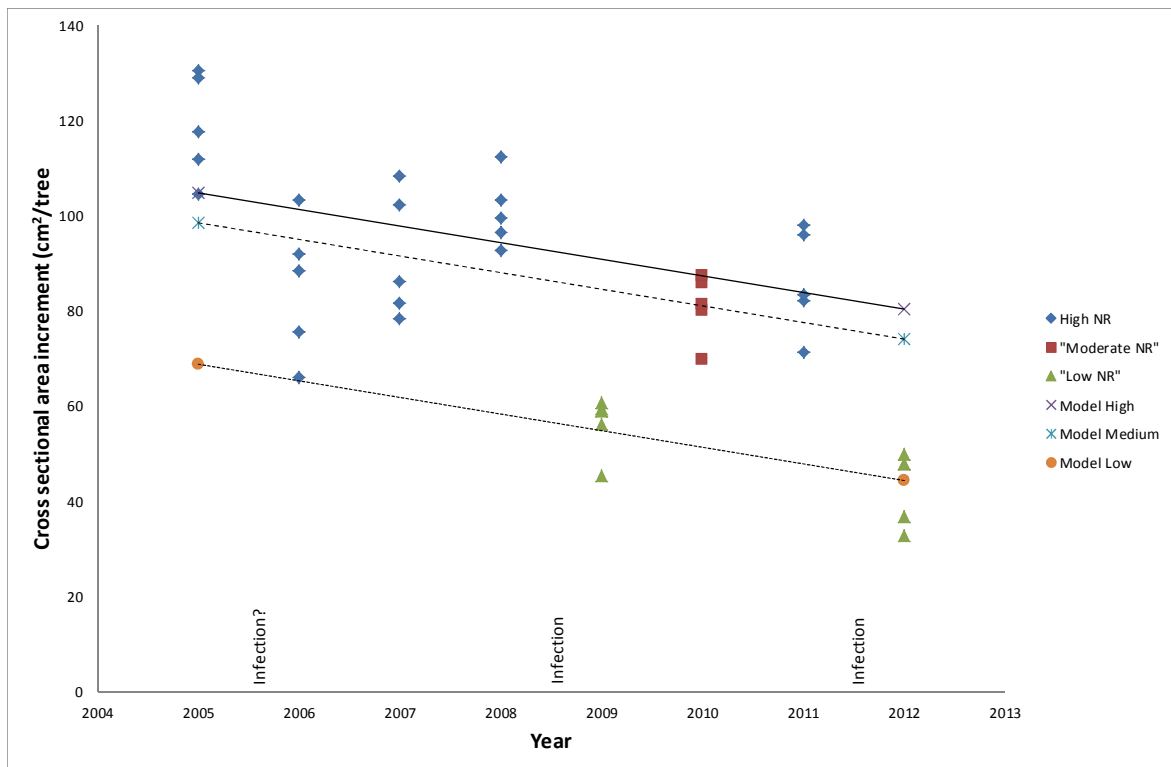
### **Growth loss per RNC event and over a rotation**

The loss in basal area increment associated with a severe RNC event will be the average of these three values (16% =  $(39 + 8 + 0)/3$ ). With severe defoliation, crown recovery following such events would take three years. The implications over a rotation can hence be estimated. RNC events have historically occurred at intervals of 3 years (since 2005?, 2008, 2011). If this infection pattern continues over a rotation, an overall loss in basal area of around 16% could be expected for such stands. The growth loss over a full rotation would be less if events occurred less frequently. Growth loss would also be less if the RNC events were less severe. RNC was severe at Wharerata Forest, so the growth loss estimates given here are likely to be at the upper end of the spectrum.



**Figure 2. Annual average cross sectional area increment of cores for all live trees per plot at Maxwell (solid lines) and Akarana plots (dashed lines), focussing on annual rings formed from 1st July 2004 to 30th June 2012**

The relationship between cross-sectional area increment and year is shown for each NR class in Figure 3. Increments were low in 2009 and 2012, following infection and needle defoliation by RNC in spring of 2008 and 2011. It is also evident that increments were low in 2006, which was a year with high rainfall, and there is a possibility that NR was low owing to a defoliation event in the spring of 2005. The analysis treats this possibility as noise. Alternatively, reclassifying 2006 and 2007 as “Low” and “Medium” retention year, respectively will improve model predictions in early years.



**Figure 3. Average cross sectional area increment of cores for all live trees per plot at Maxwell. RNC infection was known to have occurred during the Winter/Spring of 2008 and 2011, and was suspected to have occurred in 2005 (but not allowed for in the data analysis)**

If RNC was severe in the Maxwell stands in 2005, NR classes could be assigned accordingly, which will improve the estimate of growth loss owing to RNC – the analysis currently treats the drop in 2006 increments as noise within otherwise healthy stands.

### 300 Index trends

The 300 Index trends from 2003 – 2012 are shown by plot in Figure 4. AK6 had a one tree with a top-out in 2011 so the trend ceased in 2010, to avoid showing growth loss owing to wind damage. RNC in 2008 caused a slight dip in the trend lines immediately following the year of infection, but this was short lived. In general, the 300 Index values were reasonably constant over time for each plot. For example, MX1, which had the lowest needle retention scores in 2012 of any plot, had a reasonably steady 300 Index until 2008, and then decreased slightly.

Because the 300 Index is a mean annual increment, the impact of intermittent losses owing to RNC tended to be obscured. The 300 Index model could potentially be used for scenario analysis (different return times, etc.) of the impacts of RNC on growth.

### Current annual increment (CAI) in stem volume

For each plot, the CAI in stem volume was estimated annually using the 300 Index growth model from annual basal area and stocking data derived from core samples acquired in 2012, and the re-measurement in 2013. Mean top height at each age was modelled from height measurements acquired in 2012. The resulting annual stem total volume estimates (under bark) from 2004-2013 (Figure 5) were used to calculate the CAI in volume each year from 2004 to 2013 (Figure 6). Figure 6 also shows that peak defoliation events due to RNC in 2005, 2008, and 2011 were followed by low volume CAI's one year later in 2006, 2009, and 2012. Further, volume increment partially recovered two years later in 2007 (1142 mm rainfall) and 2010 (1632 mm rainfall), but not in 2013 (560 mm rainfall), even though average needle retention increased from 0.3 in 2012 to 1.2 years in 2013.



An expected improvement in volume increment probably did not occur in 2013 because of severe drought that year. Volume increment at Maxwell Road should improve markedly by 2014 provided that defoliation due to RNC does not re-occur in Spring 2013 and assuming rainfall is adequate over the coming year.

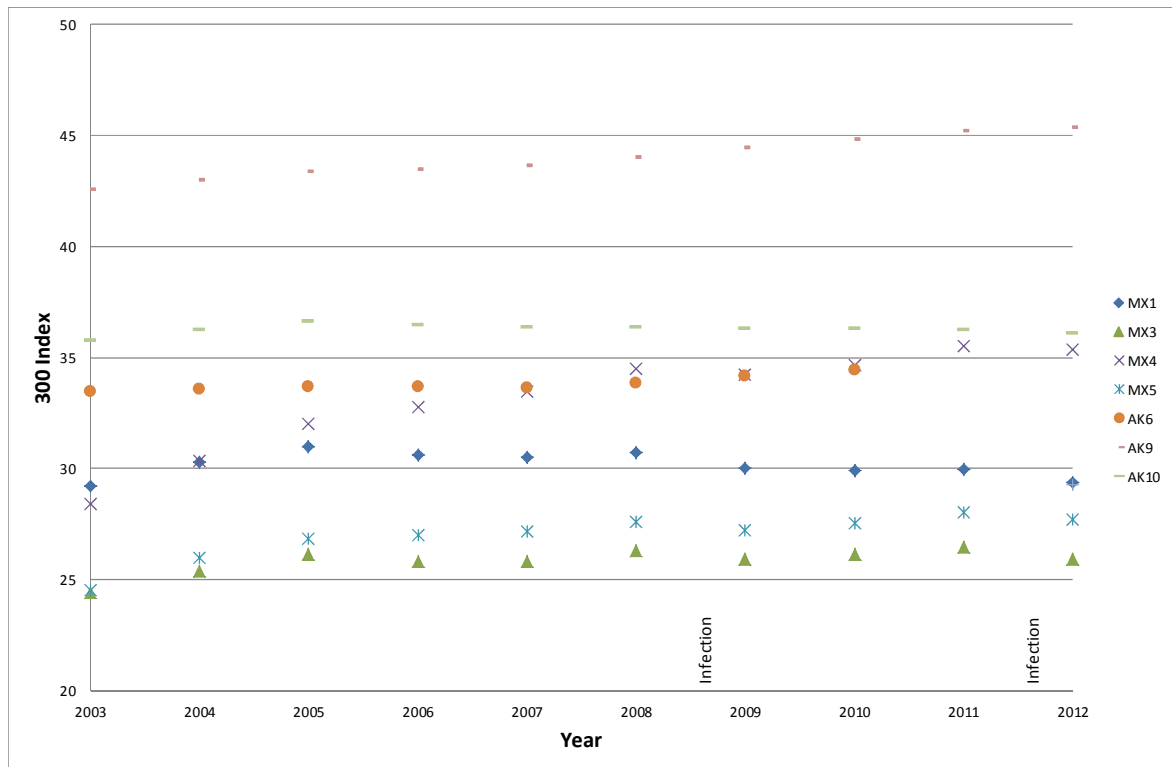


Figure 4. 300 Index values per plot from 2003 (prior to any RNC) to 2012, by which year at least two RNC infections were known to have occurred

### Effect on wood density

The analysis based on Akarana and Maxwell plots indicated that there were ephemeral effects of low needle retention owing to RNC infection on wood density attributes. Late wood (LW) density (Prob. = 0.0142) and maximum density (Prob. = 0.0004) were reduced, and consequently the density range (Prob. = 0.0062) within ring was reduced. However mean ring density was little affected, because there was a small although not statistically significant increase in earlywood density.

TABLE 3. Wood density of tree growth rings in years classified as high or low needle retention following RNC infection

| Attribute | Prob.>F | High NR | Low NR |
|-----------|---------|---------|--------|
| Density   | 0.0298  | 408     | 406    |
| LW        | 0.0142  | 532     | 506    |
| Maximum   | 0.0004  | 538     | 498    |
| Range     | 0.0062  | 206     | 175    |

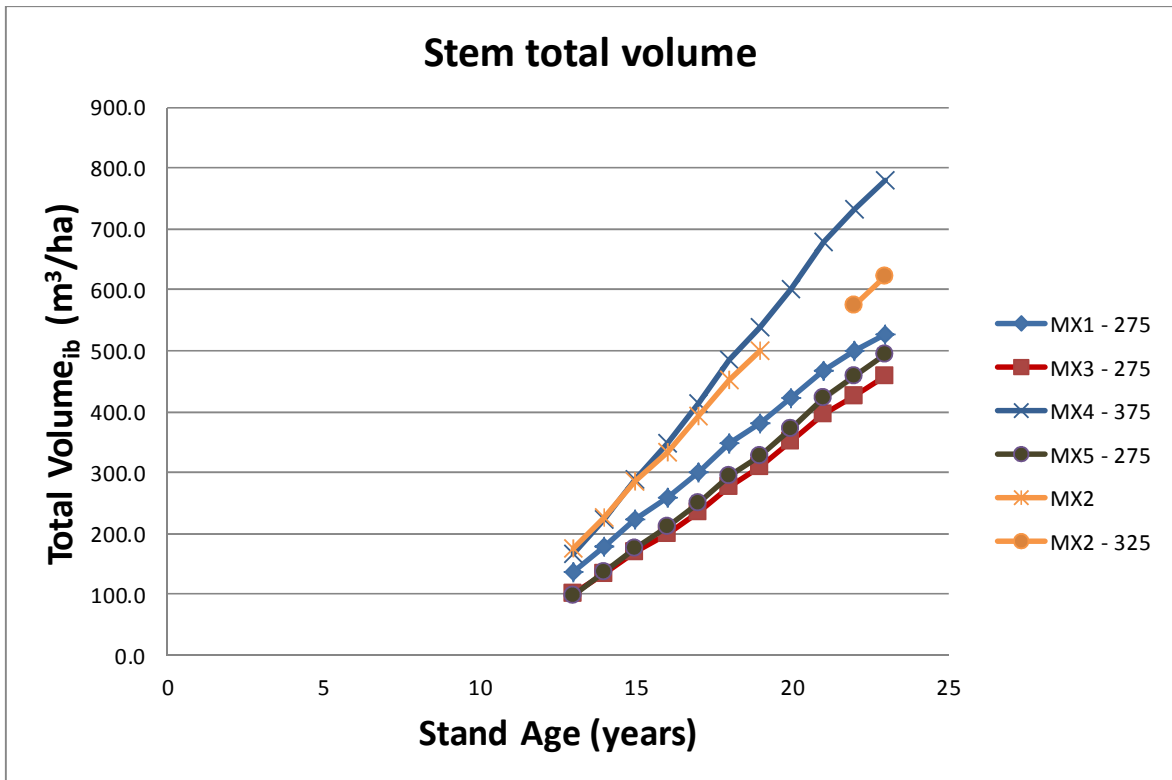


Figure 5. Stem total volume at Maxwell Road plots, calculated from basal area measurements (historically from core samples) and modelled height using the 300 Index growth model. Tree mortality occurred in one of the plots, MX2, and is shown as a data gap.

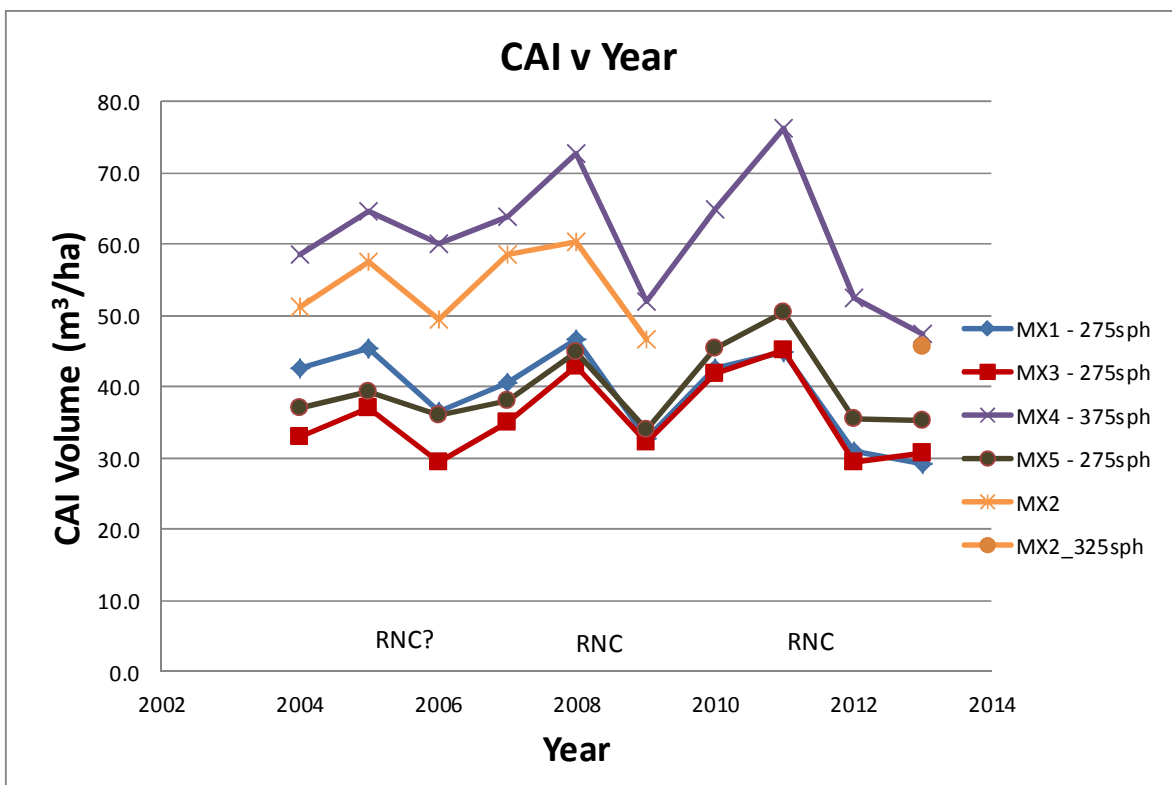


Figure 6. Current annual increment at Maxwell Road plots, based on stem volume estimates shown in Figure 5. Defoliation due to RNC occurred in 2005, 2008, and 2011.

## 4. Recommendations and Conclusions

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Severe defoliation by RNC can have an appreciable effect on current annual stem cross sectional area increment, reducing breast height increment by approximately 40% in the year immediately following infection.

It takes 3 years for the canopy to recover from a severe defoliation event, by which time stem periodic mean cross sectional area increment at breast height for the event will average around 16% less than expected in healthy stands. Volume increments based on the 2013 measurement were excluded because of exceptional drought conditions that year.

Growth loss over a full rotation will depend on the number and severity of RNC induced defoliation events.

Relationships between increment and rainfall (only two low rainfall years) were confounded at Wharerata by silvicultural and defoliation events. More sites or years of data will be required to assess the impact of rainfall on growth loss owing to RNC.

The increment core data provided clear evidence of growth loss. A similar approach using cores at a less severely defoliated site would help establish the loss in increment when RNC infection is moderate as opposed to severe.

Nevertheless, the 300 Index values only exhibited only ephemeral reductions following RNC infection years, primarily because the index averages out the effects of periodic growth loss events over the assumed 30 year life of a stand.

RNC infection has occurred every 3rd year to date. Is this because the environmental conditions were conducive only every 3rd year? If environmental conditions change such that RNC infects stands every year or every other year then the impacts on growth will be far greater than 16% per event.

Needle retention assessments at Maxwell Road plots in 2012 and 2013 showed an improvement in crown health over time, but an expected improvement in volume increment did not occur, presumably owing to severe drought over the past year.

Volume increment at Maxwell Road should increase markedly in 2014, provided that defoliation due to RNC does not re-occur in Spring 2013 and adequate rainfall occurs during the coming year.

## 5. Acknowledgements

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Andrew Costello provided stand record required to run the 300 Index model, and access to JNL forest at Wharerata.

## 6. References

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Cown, D.J.; Clement, B.C. 1983: A wood densitometer using direct scanning with X-rays. *Wood Science and Technology* 17(2): 91-99.